Roland Memisevic

Curriculum Vitae

Education

Computer Science, PhD Sep. 2003 – Mar. 2008
University of Toronto, Department of Computer Science
Adviser: Prof. Geoffrey Hinton
Thesis title: *Non-linear latent factor models for revealing structure in high-dimensional data*

Computer Science, Diplom (with distinction) Apr. 1997 – Apr. 2003
German equivalent of combined B.Sc./M.Sc.
Department of Computer Science, Bielefeld University, Bielefeld/Germany.
Advisers: Prof. Helge Ritter and Dr. Peter Meinicke
Thesis title: *Unsupervised Kernel Regression for Nonlinear Dimensionality Reduction*

Professional Experience

Qualcomm AI Research since July 2021
Sr. Director

Twenty Billion Neurons June 2016 – June 2021
Co-founder, Chief Scientist (2016-2018), CEO (2018-2021)

Department of Computer Science and Operations Research, University of Montreal, Canada Sep 2012 – May 2020
(adjunct 2016-2020)
Assistant/adjunct Professor at MILA (Montreal Institute for Learning Algorithms)

Department of Computer Science, University of Frankfurt Jan. 2011 – Aug. 2012
Juniorprofessor

Department of Computer Science, ETH Zürich, Switzerland Jan. 2010 – Dec. 2010
Postdoctoral fellow

PNYLab LLC, Princeton, USA Jul. 2008 – Dec. 2009
Research scientist

Department of Computer Science, University of Toronto, Canada Jan. 2008 – Jun. 2008
Postdoctoral fellow

Microsoft Research, Redmond, USA Jun. 2007 – Sep. 2007
Research intern

Department of Computer Science, Bielefeld University, Germany Apr. 2003 – Oct. 2003
Research assistant

Publications

Liu, L., Hou, Q., Jian, Y., Pourreza, R., Ghavamzadeh, M., Memisevic, R., Qin, Y., Cai, H. 2026
Alleviating Hallucination in Reasoning Tasks with Training-Free Uncertainty-Guided Steering
Neurips 2026

Madan, P., Haresh, S., Ebrahimi, R., Panchal, S., Bhattacharyya, A., (additional authors), Memisevic, R. 2026
On Locality and Length Generalization in Visual Reasoning
ECCV 2026

Haresh, S., Dijkman, D., Bhattacharyya, A., Memisevic, R. 2026
Notes-to-Self: Scratchpad Augmented VLAs for Memory Dependent Manipulation Tasks
ICRA 2026

Ebrahimi, R., Defferrard, M., Panchal, S., Memisevic, R. 2026
On the "Induction Bias" in Sequence Models
ICML 2026

Liu, L., Pourreza, R., Jian, Y., Qin, Y., Memisevic, R. 2026
From out-of-distribution detection to hallucination detection: A geometric view
ICML 2026

Pourreza, R., Dagli, R., Bhattacharyya, A., Panchal, S., Berger, G., Memisevic, R. 2026
Can Vision-Language Models Answer Face to Face Questions in the Real-World?
ICLR 2026

Liu, L., Pourreza, R., Panchal, S., Bhattacharyya, A., Jian, Y., Qin, Y., (additional authors), Memisevic, R. 2026
Enhancing hallucination detection through noise injection
ICLR 2026

Bhattacharyya, A., Mahajan, S., Haresh, S., Yasarla, R., Pourreza, R., Liu, L., (additional authors), Memisevic, R. 2026
Streaming Interventions: Can Video Large Language Models Correct Mistakes as They Occur?
Neurips 2026

Bhattacharyya, A., Xu, B., (5 additional authors), Sigal, L., Memisevic, R. 2025
Can Multi-Modal LLMs Provide Live Step-by-Step Task Guidance?
Neurips 2025

Ebrahimi, R., Memisevic, R. 2025
Revisiting Bi-Linear State Transitions in Recurrent Neural Networks
Neurips 2025

Khisti, A., Ebrahimi, R., Dbouk, H., Behboodi, A., Memisevic, R., Louizos C. 2025
Multi-Draft Speculative Sampling: Canonical Decomposition and Theoretical Limits
ICLR 2025

Phan, B., Ebrahimi, R., Haresh, S., Memisevic, R. 2025
Delayed Attention Training Improves Length Generalization in Transformer–RNN Hybrids
Neurips Workshop on what transformers can(t) do

Rainone, C., Bakker, T., Memisevic, R. 2025
Replacing thinking with tool usage enables reasoning in small language models
arXiv preprint

Haresh, S. Dijkman, D., Bhattacharyya, A., Memisevic, R. 2024
ClevrSkills: Compositional Language and Visual Reasoning in Robotics
Neurips 2024

Ebrahimi, R., Panchal, S., Memisevic, R. 2024
Your context is not an array: Unveiling random access limitations in transformers
COLM 2024

Panchal, S., Bhattacharyya, A., Berger, G., (8 additional authors), Memisevic, R. 2024
Live Fitness Coaching as a Testbed for Situated Interaction
Neurips 2024

Dagli, R., Berger, G., Materzysnka, J., Bax. I., Memisevic, R. 2024
AirLetters: An Open Video Dataset of Characters Drawn in the Air
ECCV Hands workshop 2024

Bhattacharyya, A., Panchal S., Pourreza, R., Lee, M., Madan, P, Memisevic, R. 2024
Look, Remember and Reason: Grounded Reasoning in Videos with Language Models
ICLR 2024

Ling, Z., Fang, X., Li, X., Mu, T., Lee, Memisevic, R., Su, H. 2023
Deductive Verification of Chain-of-Thought Reasoning
Neurips 2023

Pourreza, R., Bhattacharyya, A., Panchal, S., Lee, M., Madan, P., Memisevic, R. 2023
Painter: Teaching Auto-regressive Language Models to Draw Sketches
ICCV Workshops 2023

Memisevic, R., Panchal, S., Lee, M. 2022
Decision making as language generation
Neurips workshop on Foundation Models for Decision Making

Mercier, A., Berger, G., Panchal, S., Letsch, F., Boehm, C., Bax, I. Memisevic, R. 2020
Does action recognition need pose estimation?
Medium

Materzynska, J., Berger G., Bax, I., Memisevic, R. 2019
The jester dataset: A large-scale video dataset of human gestures
ICCV workshop HANDS 2019

Mahdisoltani, F., Berger, G., Gharbieh, W., Memisevic, R., Fleet, D. 2018
The more fine-grained, the better for transfer learning
NeurIPS Workshop on Modeling and Decision-Making in the Spatiotemporal Domain

Goyal, R., Mahdisoltani, F., Berger, G., Gharbieh, W., Bax, I., Memisevic R. 2018
Evaluating visual "common sense" using fine-grained classification and captioning tasks
ICLR workshop

Ebrahimi Kahou, S., Michalski, V., Memisevic, R., Pal, C., Vincent, P. 2017
RATM: Recurrent attentive tracking model
CVPR workshop BNMR

Goyal, R., Kahou, S., Michalski, V., (9 additional authors), Memisevic, R. 2017
The "something something" video database for learning and evaluating visual common sense
International Conference on Computer Vision (ICCV 2017)

Berger, G., Memisevic, R. 2017
Incorporating long-range consistency in CNN-based texture generation
International Conference on Learning Representations (ICLR 2017)

Lin, Z., Courbariaux M., Memisevic, R., Bengio Y. 2016
Neural Networks with Few Multiplications
International Conference on Learning Representations (ICLR 2016)

Krueger, D., Memisevic, R. 2016
Regularizing RNNs by Stabilizing Activations
International Conference on Learning Representations (ICLR 2016)

Im, D., Belghazi, M., Memisevic, R. 2016
Conservativeness of Untied Auto-Encoders
Thirtieth AAAI Conference on Artificial Intelligence (AAAI 2016)

de Vries, H., Memisevic, R., Courville, A. 2016
Deep Learning Vector Quantization
European Symposium on Artificial Neural Networks (ESANN 2016)

Zhang,S., Wu,Y., Che,T., Lin,Z., Memisevic,R., Salakhutdinov,R., Bengio,Y. 2016
Architectural Complexity Measures of Recurrent Neural Networks
Neural Information Processing Systems (NIPS 2016).

Kamyshanska, H., Memisevic, R. 2015
The potential energy of an autoencoder
IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI).

Konda, K., Memisevic, R., Krueger, D. 2015
Zero-bias autoencoders and the benefits of co-adapting features
International Conference on Learning Representations (ICLR 2015).

Konda, K., Memisevic, R. 2015
Learning visual odometry with a convolutional network
International Conference on Computer Vision Theory and Applications (VISAPP 2015).

Jean, S., Cho, K., Memisevic, R., Bengio, Y. 2015
On using very large target vocabulary for neural machine translation
Annual meeting of the Association for Computational Linguistics (ACL 2015).

Konda, K., Chandrasekariah, P., Memisevic, R., Triesch, J. 2015
Real-time activity recognition via deep learning of motion features
European Symposium on Artificial Neural Networks (ESANN2015)

Michalski V., Memisevic R., Konda K. 2014
Modeling Deep Temporal Dependencies with Recurrent "Grammar Cells"
Neural Information Processing Systems (NIPS 2014).

Konda K., Memisevic, R., Michalski V. 2014
Learning to encode motion using spatio-temporal synchrony
International Conference on Representation Learning (ICLR 2014).

Konda K., Memisevic R. 2014
A unified approach to learning depth and motion features
Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP 2014).

Memisevic, R. 2013
Learning to relate images
IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI) Special issue on Learning Deep Architectures 2013.

Kamyshanska, H., Memisevic, R. 2013
On autoencoder scoring
International Conference on Machine Learning (ICML 2013).

Memisevic, R., Exarchakis, G. 2013
Learning invariant features by harnessing the aperture problem
International Conference on Machine Learning (ICML 2013).

Bauer, F., Memisevic, R. 2013
Feature grouping from spatially constrained multiplicative interaction
International Conference on Representation Learning (ICLR 2013).

Memisevic, R., Sigal L., Fleet, D. 2012
Shared Kernel Information Embedding for Discriminative Inference.
IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI).

Memisevic, R. 2012
On multiview feature learning
International Conference on Machine Learning (ICML 2012).

Memisevic, R. 2012
On spatio-temporal feature learning
The Learning Workshop, Snowbird.

Memisevic, R. 2011
Gradient-based learning of higher-order image features.
International Conference on Computer Vision (ICCV 2011).

Susskind, J., Memisevic, R., Hinton, G., Pollefeys M. 2011
Modeling the joint density of two images under a variety of transformations.
International Conference on Computer Vision and Pattern Recognition (CVPR 2011).

Memisevic, R. 2011
On spatio-temporal sparse coding: Analysis and an algorithm.
NIPS workshop on deep learning and unsupervised feature learning.

Memisevic, R., Conrad C. 2011
Stereopsis via deep learning.
NIPS workshop on deep learning and unsupervised feature learning.

Memisevic, R., Hinton, G. 2010
Learning to Represent Spatial Transformations with Factored Higher-Order Boltzmann Machines.
Neural Computation. Vol. 22, No. 6: 1473-1492.

Memisevic, R., Zach, C., Hinton, G., Pollefeys M. 2010
Gated Softmax Classification.
Neural Information Processing Systems (NIPS 2010).

Sigal, L., Memisevic, R., Fleet, D. 2009
Shared Kernel Information Embedding for Discriminative Inference.
International Conference on Computer Vision and Pattern Recognition (CVPR) 2009.

Memisevic, R., Hinton, G. E. 2007
Unsupervised learning of image transformations.
International Conference on Computer Vision and Pattern Recognition (CVPR) 2007.

Samulowitz, H., Memisevic, R. 2007
Learning to solve QBF.
Twenty-Second Conference on Artificial Intelligence (AAAI) 2007.

Memisevic, R. 2006
Kernel Information Embeddings.
International Conference on Machine Learning (ICML) 2006.

Memisevic, R. 2006
Propagating Errors and Beliefs for Large Scale Nonlinear Structure Prediction.
North East Student Colloquium on Artificial Intelligence. Ithaca, NY.

Meinicke, P., Klanke, S., Memisevic, R., Ritter, H. 2005
Principal Surfaces from Unsupervised Kernel Regression.
IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), Sep. 2005.

Memisevic, R. and Hinton, G. E. 2005
Improving dimensionality reduction with spectral gradient descent.
Neural Networks, 18, pp 702-710.

Memisevic, R., Srebro N., Roweis, S. 2005
Exploring the regularization path for adaptive Gaussian kernel SVMs.
NIPS Workshop on the Accuracy-Regularization Frontier.

Memisevic, R. 2005
Dual Optimization of Conditional Probability Models.
NIPS Workshop on Kernel Methods for Structured Domains.

Memisevic, R. and Hinton, G. E. 2005
Embedding via clustering: Using spectral information to guide dimensionality reduction. International Joint Conference on Neural Networks 2005.

Memisevic, R. and Hinton, G. E. 2005
Multiple Relational Embedding.
Neural Information Processing Systems (NIPS) 2005.

Theses

Memisevic, R. 2008
Non-linear latent factor models for revealing structure in high-dimensional data.
PhD Thesis, Department of Computer Science, University of Toronto.

Memisevic, R. 2003
Unsupervised Kernel Regression for Nonlinear Dimensionality Reduction.
Master's thesis, Department of Computer Science, Bielefeld University.

Selected talks

Workshop on NLP in the Era of Generative AI, Cognitive Sciences, and Societal Transformation
Invited talk "End-to-end learning of camera-based, situated interactions" MILA, Montreal 2024

Qualcomm webinar "Integrating senses: how AI is learning to see, hear, and interact" Sep. 2024

ISCAS 2017 Invited talk Baltimore May 2017

Re-Work Deep Learning Summit 2017 Invited talk San Francisco Jan. 2017

MIT Deep Learning workshop 2016 Invited talk Boston Jun. 2016

ICISP 2016 Keynote Trois Rivieres Jun. 2016

Re-Work Deep Learning Summit 2016 Invited talk San Francisco Jan. 2016

Data science meetup Berlin Invited talk Berlin Nov. 2015

University of Toronto Invited seminar, CS department Toronto Nov. 2015

KLA-Tencor Invited talk Monterey Oct. 2015

Google Invited talk Mountain View Oct. 2015

ICML 2015 workshop on Deep Learning Invited talk Lille July 2015

Universite Laval Invited talk Dept. for Comp. Sci. and Software Engineering Quebec Apr. 2015

Data science day Invited talk Berlin Oct. 2014

ICLR 2014 Keynote Banff Apr. 2014

Max-Planck Institute for Biological Cybernetics Invited talk Tuebingen Aug. 2011

Learning the meaning of "the same", University of Basel. Basel Aug. 2010

How to train a mixture of 100.000.000.000.000.000 classifiers., Univ. of Toronto. May 2010

Learning to encode data transformations., ETH Zürich. Sep. 2009

Extrahieren verborgener Strukturen mit latenten Variablen., TU Darmstadt. Sep. 2009

Learning with factorized three-way interactions., CIFAR workshop on inter-neurons. Apr. 2008

Learning the allowable transformations of simple shapes using 3-way interactions., CIFAR workshop on 'Achieving perceptual invariance'. Dec. 2007

Building a compact multi-resolution index structure., Microsoft Research. Sep. 2007

Learning with conditional latent variables., Yahoo! Applied Research Montreal. June 2007

Learning with conditional latent variables., TradeLink Holdings, Chicago. May 2007

Unsupervised learning of image transformations., CIFAR Summer School. Aug. 2006

Kernel information embeddings, University of Toronto. Sep. 2005

Kernel information embeddings. CIFAR Workshop on Manifold Learning. Apr. 2005

Grants and Awards

FQRNT Team Research Project: Unsupervised deep learning of changes in facial expression 2015
$\approx \$150,000$ CAD four three years. Team leader, with co-PIs Yoshua Bengio, Pascal Vincent, Aaron Courville.

DARPA seedling project 2015
$\approx \$180,000$ USD for one year for studying Deep Learning based time series models (PI) jointly with Yoshua Bengio (co-PI). Part of a larger project with SRI and University of Guelph.

Facebook 2014
$\$25,000$ USD, gift

Google faculty research award 2013
$\$40,800$ USD for studying phase-sensitive feature hierarchies.

NSERC Discovery Grant 2013
$\$100,000$ CAD ($\$20,000$ per year for five years 2013–2018) Research Grant.

NSERC Strategic Grant 2013
$\$575,739$ CAD ($\$191,913$ per year for three years) Strategic Project Grant in collaboration with Ubisoft (jointly with Yoshua Bengio, Pascal Vincent and Aaron Courville)

Government of Canada Award 2003
Funding for the first two years of my PhD program at Toronto ($>$ \$50.000 CAD in total)

Professional activities

Regular reviewer and/or PC member for conferences ICLR, NIPS, ICML, CVPR, ICCV, ECCV, DAGM

Reviewer for Swiss National Science Foundation, Journal IEEE Transactions on Pattern Analysis and Machine Intelligence (PAMI), Journal of Machine Learning Research (JMLR), Journal Image and Vision Computing (IMAVIS), Journal Constraint, Journal Machine Learning, IEEE Transactions on Image Processing, International Journal of Adaptive Control and Signal Processing.

Co-organizer for CRM/CIFAR Summer School on Deep Learning in Montreal 2015 (with Yoshua Bengio, Yann LeCun)

Co-organizer for NIPS 2014 workshop on Deep Learning and Representation Learning (with Adam Coates, Yoshua Bengio, Andrew Ng)

Invited tutorial on Deep Learning at *Hotchips* (Cupertino, CA) 2015

Invited tutorial on Deep Learning in Image Processing and Vision at International Conference on Image Processing *ICIP* (Quebec City) 2015

Invited tutorial at Training Workshop on Deep Architectures in Vision and NL (Wroclaw, Poland) 2015

Invited tutorial at the *CIFAR Summer School Neural Computation and Adaptive Perception (NCAP)* Toronto 2013

Tutorial at CVPR 2012: *Multiview feature learning*

Invited tutorial at the *IPAM Graduate Summer School: Deep Learning, Feature Learning* (UCLA) 2012

Invited tutorial at DAGM 2011: *Higher-order feature learning: Building a computer vision "swiss army knife"*

Graduate student supervision

Zhouhan Lin (PhD student at University of Montreal, from 2014)

Vincent Michalski (PhD student at University of Montreal, from 2015)

Harm de Vries (PhD student at University of Montreal, from 2014) jointly with Aaron Courville

Kyle Kastner (MSc student at University of Montreal, from 2014) jointly with Yoshua Bengio

David Krueger (MSc student at University of Montreal, from 2013) jointly with Yoshua Bengio

Kishore Konda (University of Frankfurt) PhD received 2015

Sebastien Jean (University of Montreal) MSc received 2015

Vincent Michalski (University of Frankfurt) MSc received 2014

Irina Shelegeda (University of Frankfurt) MSc received 2014

Hanna Kamyshanska (University of Frankfurt) MSc received 2013

Felix Bauer (University of Frankfurt) MSc received 2012

Teaching

Instructor, *Machine Learning for Vision (IFT 6268)* Winter 2016
University of Montreal

Instructor, *Databases (IFT2821)* Winter 2015
University of Montreal

Instructor, *Machine Learning for Vision (IFT 6268)* Winter 2015
University of Montreal

Instructor, *Fundamentals of Machine Learning* Fall 2014
University of Montreal

Instructor, *Machine Learning for Vision (IFT 6268)* Fall 2013
University of Montreal

Instructor, *Topics in artificial intelligence (IFT 6085): Visual Feature Learning* Winter 2013
University of Montreal

Instructor, *Machine Learning* Winter 2011/12
University of Frankfurt

Instructor, *Digital Image Processing* Spring 2011
University of Frankfurt

Teaching Assistant, *Introduction to Neural Networks and Machine Learning (CSC 321)* Spring 06/07
University of Toronto

Teaching Assistant, *Software Design* (CSC 207) , University of Toronto Summer 2006

Teaching Assistant, *Machine Learning* (CSC 2515), University of Toronto Fall 2005 and 2006

Teaching Assistant, *Theoretical Computer Science*, Bielefeld University Summer 2000

Teaching Assistant, *Methods of Artificial Intelligence*, Bielefeld University Winter 1999/2000